

Se diseñó y desarrolló una solución automatizada para la generación de reportes de trámites, mediante la construcción de un script en Python que permite procesar de manera sistemática la información consolidada y generar reportes individuales por municipio, a partir de una plantilla estructurada previamente definida.

Generar reportes por municipio

Subtask:

For each unique municipality, populate the Word document template with its specific data.

```
import os
from docx import Document

# Create an output folder if it doesn't exist
output_folder = 'reports_word'
if not os.path.exists(output_folder):
    os.makedirs(output_folder)
    print(f"Created output folder: {output_folder}")

# Function to replace placeholder text in paragraphs while preserving formatting
def replace_placeholder_in_paragraph(paragraph, placeholder, replacement):
    if placeholder in paragraph.text:
        for run in paragraph.runs:
            if placeholder in run.text:
                run.text = run.text.replace(placeholder, replacement)
                # Assuming placeholder only appears once per run or paragraph
                return True
    return False

# Iterate through each unique municipality to generate a report
for municipality in unique_municipalities:
    print(f"Generating report for: {municipality}")

    # Step 2: Create a new Document object by loading the template file
    doc = Document(template_path)

    # Step 3: Retrieve the specific DataFrame for the current municipality
    current_municipality_df = municipality_data[municipality]

    # Step 4: Iterate through all paragraphs and replace the municipality name placeholder
    for paragraph in doc.paragraphs:
        replace_placeholder_in_paragraph(paragraph, '{{MUNICIPIO_NAME}}', municipality)
        replace_placeholder_in_paragraph(paragraph, '{{MUNICIPIO}}', municipality)
```